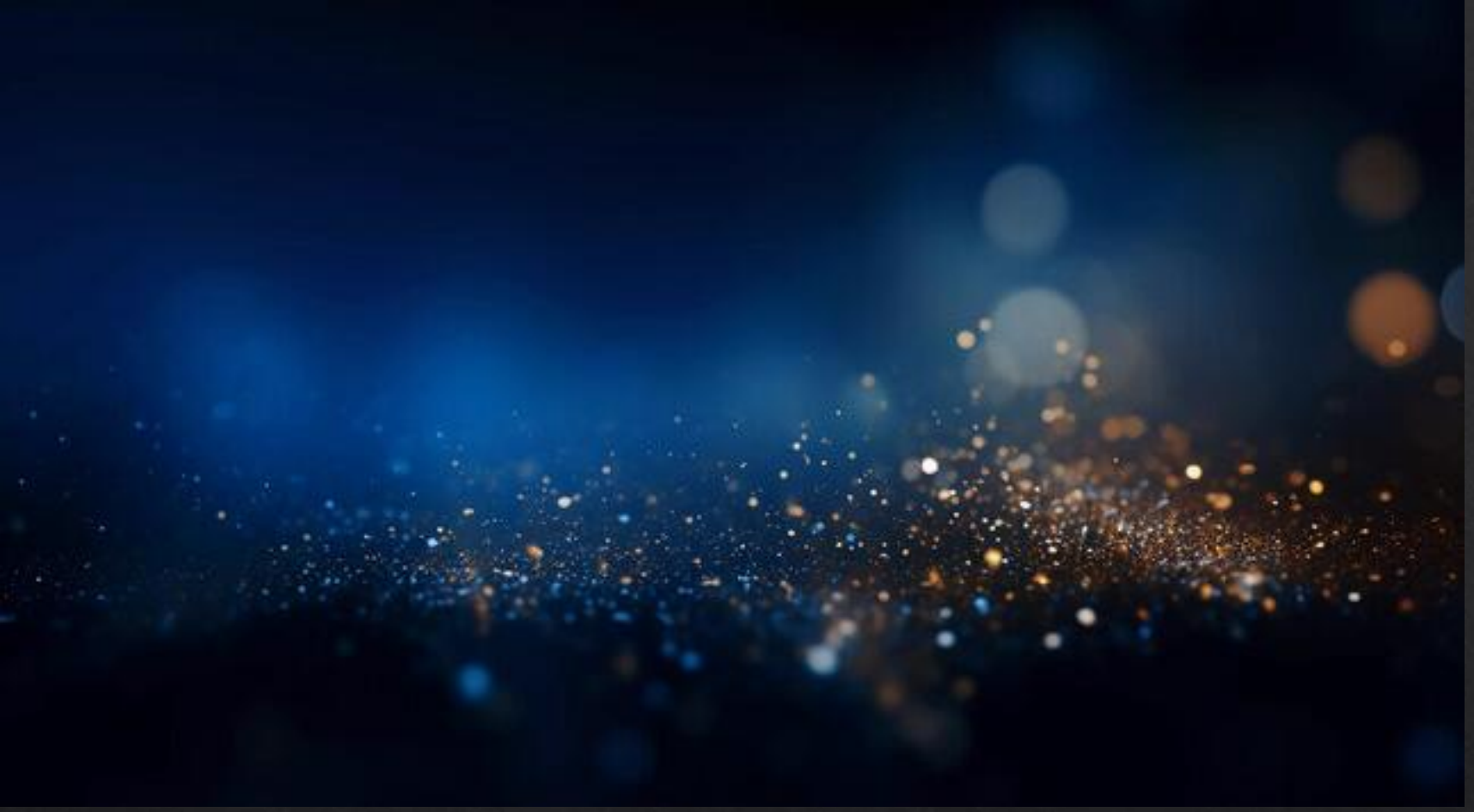


MA416

Lecture 16 – October 28, 2025

Common Non-Parametric Methods

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Sign Test for Percentiles

Suppose we desire to test $H_0: \mu = \mu_0$ but $S \subseteq P$ is not associated to a normally distributed population. If P was to be normal, then $\mu = M_x = x^L_{50\%} = x^U_{50\%}$. If P was not normal, then $\exists p \in (0,1)$ such that $\mu = x_{p\%}$.

Example: Suppose that $S = \{1, 5, 8, 9, 10, 11, 16, 19, 23\}$ and someone claims that:

$$H_0: x^L_{50\%} = 9$$

This is equivalent to say claiming that:

$$\mathbb{P}_0[X \leq 9] \coloneqq p^{(0)}_{\leq 9} = 0.50$$

For our sample, we have:

$$\hat{p}_{\leq 9} = \frac{4}{9} \approx 0.444$$

Question: Is $\hat{p}_{\leq 9}$ significantly different from $p^{(0)}_{\leq 9}$?

Sign Test for Percentiles

$$\hat{p}_{\leq 9} = \frac{4}{9} \approx 0.444 \qquad vs. \qquad p_{\leq 9}^{(0)} = 0.50$$

This then turns this into a one-sample proportion test. Recall that if the observations within the sample are independent, then if $\hat{p} = \frac{x_s}{n}$, then x_s follows a binomial distribution with parameters n, p. From this, one can show that:

$$\mathbb{E}(\hat{p}) = p \qquad \mathbb{V}(\hat{p}) = \frac{p(1-p)}{n} \approx^{wald} \frac{\hat{p}(1-\hat{p})}{n} \quad \rightarrow \quad \text{SE}(\hat{p}) \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

By the De'Moivre-Laplace theorem, if $n \rightarrow \infty$, the distribution of $\frac{\hat{p}-p}{\text{SE}(\hat{p})} \rightarrow^d Z$. This then gives us the ability to do confidence intervals, test statistics, and p-values in the usual way under the assumptions of independent observations, and if Z is to be used, then sample size must be mandated to be large.

Rank-Based Testing

Suppose that you have two sets for which one desires to test the claim that:

$$H_0: \mu_1 = \mu_2$$

Note if the distribution of P_1 equals that of P_2, then their parameters (and also their means) should roughly be close to one another.

Example: Let $S_1 = \{1, 4, 7, 7, 9, 10\}$ and $S_2 = \{11, 16, 20, 20, 20\}$.

One should be able to infer from the values above that there is adequate evidence to support $H_a: \mu_1 \neq \mu_2$. If one were to look at the ranks of all of the values together (from least to greatest):

Sample	1	1	1	1	1	1	2	2	2	2	2
Value	1	4	7	7	9	10	11	16	20	20	20
Rank*	1	2	3	4	5	6	7	8	9	10	11
Rank	1	2	3.5	3.5	5	6	7	8	10	10	10

Note that if there are ties, one should assign a rank equal to that of the average of the default ranks they would be assigned to. For the example above, we then have:

$$R_1 = \{1, 2, 3.5, 3.5, 5, 6\} \quad \text{and} \quad R_2 = \{7, 8, 10, 10, 10\}$$

Rank-Based Testing

R-Implementation

```
S1 = c(1, 4, 7, 7, 9, 10)
S2 = c(11, 16, 20, 20, 20)

L = list('s1' = S1, 's2' = S2)
nj = sapply(L, length)
n = sum(nj)
```

```
Sall = unlist(L)
Sall

idx = rep(names(L), times=nj)
idx
```

```
> Sall
s11 s12 s13 s14 s15 s16 s21 s22 s23 s24 s25
  1   4   7   7   9  10  11  16  20  20  20
> idx
 [1] "s1" "s1" "s1" "s1" "s1" "s1" "s2" "s2" "s2" "s2" "s2"
```

```
Rall = rank(Sall)
Rall

Rj = split(Rall, idx)
Rj
```

```
> Rall
 s11  s12  s13  s14  s15  s16  s21  s22  s23  s24  s25
 1.0  2.0  3.5  3.5  5.0  6.0  7.0  8.0 10.0 10.0 10.0
> Rj
$s1
s11 s12 s13 s14 s15 s16
1.0 2.0 3.5 3.5 5.0 6.0

$s2
s21 s22 s23 s24 s25
  7   8  10  10  10
```

Two-Sample Rank-Based Testing

Once you have your ranks, R_j, for each of your groups, $j = 1,2,\dots,g$, then we can define some statistical metrics on those ranks. Suppose that $|R_j| = n_j$. The first metric is the total rank:

$$T_j = \sum_{i=1}^{n_j} (R_j)_i$$

Note that $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$, which gives the value of $(T_1 + T_2 + \cdots + T_g)$.

Testing Strategy: (Conover-Iman) "If R_i is significantly different than R_j, then T_i will likely be significantly different than T_j".

$$\mathrm{CI}_{stat} = \frac{T_1 - T_2}{\sqrt{\frac{n(n+1)}{12} \cdot \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim T_{n_1+n_2-2}$$

Not very reliable when $|n_1 - n_2| \gg 0$.

Two-Sample Rank-Based Testing

$$\mathrm{CI}_{stat} = \frac{T_1 - T_2}{\sqrt{\frac{n(n+1)}{12} \cdot \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim T_{n_1+n_2-2}$$

Define the "partial U" test statistics to be:

$$U_j = T_j - \frac{n_j(n_j+1)}{2}$$

Theorem: $U_j \in [0, n_1 n_2], U_1 + U_2 = n_1 n_2$, and $U_j \sim W_{(n_1,n_2)}$ (Wilcoxon) distribution.

Testing Strategy: (Mann-Whitney) "If R_i is significantly different than R_j, then U_i will be significantly different than U_j".

$$\mathrm{MW}_{stat} := \min\{U_1, U_2\}$$

This is commonly treated as a left-tailed (or right-tailed) test.

```
MWcrit = qwilcox(0.05, nj[1], nj[2], lower.tail=T)
```

Multiple-Sample Rank-Based Testing

If $g \geq 3$, we define:

$$\bar{R}_j = \frac{T_j}{n_j} \qquad \text{and} \qquad \bar{\bar{R}} = \frac{T_1 + T_2 + \cdots + T_g}{n} = \frac{n+1}{2}$$

and define the Kruskal-Wallis test statistics to be:

$$\mathrm{KW}^{o}_{stat} \coloneqq (n-1) \cdot \frac{\sum_{j=1}^{g} \left[n_j \left(\bar{R}_j - \bar{\bar{R}} \right)^2 \right]}{\sum_{j=1}^{g} \sum_{i=1}^{n_j} \left(R_{ij} - \bar{\bar{R}} \right)^2}$$

or if in the case that no ties are present in $S_1 \cup S_2 \cup \ldots \cup S_g$, we have:

$$\mathrm{KW}^{o}_{stat} = \frac{12}{n(n+1)} \cdot \sum_{j=1}^{g} \left[n_j \left(\bar{R}_j - \bar{\bar{R}} \right)^2 \right]$$

Both of which follow a χ^2_{g-1} if the within/between sample independence assumptions are met. We also have:

$$\mathrm{KW}^{m}_{stat} \coloneqq (n-1) \cdot \frac{\sum_{j=1}^{g} \left[n_j \left(\bar{R}_j - \bar{\bar{R}} \right)^2 \right]}{\sum_{j=1}^{g} \sum_{i=1}^{n_j} \left(R_{ij} - \bar{R}_j \right)^2}$$

which will follow a $F_{(g-1,n-g)}$ and is typically better for smaller sample sizes and is asymptotically χ^2_{g-1} for large n.

Overall Summary of N.P. Statistics So Far…

One-Sample Tests on "Means"

- Sign Tests on Percentiles

Two-Sample Tests on "Means"

- Conover-Iman
- Mann-Whitney

Multiple-Sample Tests on "Means"

- Kruskal-Wallis
- Pairwise-comparisons

Question: What about one-sample claims on "variances"?

One framework one could do is to analyze the distance between specific percentiles of interest.

Example: If $H_0: \sigma = 1$ and $X \sim N(0,1)$, then this is approximately equivalent to:

$$H_0: x^L_{16\%} = -1 \text{ and } x^L_{84\%} = 1$$

```
> pnorm(-1, 0, 1, lower.tail=T)
[1] 0.1586553
> pnorm(1, 0, 1, lower.tail=T)
[1] 0.8413447
```

Each can be tested independently and then concluded.

Multiple-Sample "Variance" Testing

The Conover test for non-parametric dispersion is typically the go-to for $g \geq 2$. We first start off with our data in $S_1, S_2, \ldots, S_g$. From here, we center our data via metric of our choosing*:

$$S_j^c \coloneqq |S_j - c_j|, \qquad c_j \coloneqq^* \text{median}(S_j)$$

From here, we calculate the associated rankings on S_j^c.

```
mydata = read.csv('CompanyPayroll_9.25.csv')

Sj = split(mydata$Efficiency_Factor_Obs, mydata$Job_Rank)
idx = rep(names(Sj), times=sapply(Sj, length))

Sjc = sapply(Sj, function(x) x-median(x))

Sjcall = unlist(Sjc)
Rall = rank(Sjcall)
Rj = split(Rall, idx)
```

We define:

$$\text{Co}_{stat} \coloneqq \left(\frac{n-g}{g-1}\right) \cdot \frac{\sum_{j=1}^{g}\left[n_j\left(\bar{R}_j - \bar{\bar{R}}\right)^2\right]}{\sum_{j=1}^{g}\sum_{i=1}^{n_j}\left(R_{ij} - \bar{R}_j\right)^2}$$

which is approximately $F_{(g-1,n-g)}$ provided assumptions are met. Note that there is a "Conover squared-ranks" test in the literature; these aren't the same, but still analyzes the S_j^c values.